

Session 21 Quiz - AI Safety, Misuse & Governance

Track B | Grades 9–12 | 20 Questions

Q1 [Type: MCQ | Difficulty: Easy | QID : 54942]

AI _____ happens when a person deliberately uses an AI system to cause harm - for example, using generative AI to flood the internet with disinformation or creating fake audio to commit fraud.

- A. Safety failure
- B. Bias
- C. Misuse
- D. Inference

Correct Answer: C

Explanation: Misuse is defined by deliberate human intent to cause harm. AI itself has no intentions - it is a tool. The technology amplifies whatever intent the human has, for better or worse.

Q2 [Type: MCQ | Difficulty: Easy | QID : 54955]

A _____ is a piece of media - a video, image, or audio recording - that has been artificially generated or manipulated using AI to make it look or sound like a real person said or did something they never actually did.

- A. Deepfake
- B. Transcript
- C. Token
- D. Hallucination

Correct Answer: A

Explanation: The word combines "deep learning" with "fake." Deepfakes threaten political stability, personal reputations, and even basic trust in genuine evidence - because once anyone can claim a real video is "just a deepfake," even truthful media becomes questionable.

Q3 [Type: MCQ | Difficulty: Medium | QID : 54957]

The _____ is widely considered the world's first comprehensive AI law. It classifies AI applications by risk level - from minimal (such as spam filters) to unacceptable (such as social scoring by governments) - and requires strict testing for high-risk systems before deployment.

- A. GDPR
- B. Global Partnership on AI
- C. Blueprint for an AI Bill of Rights
- D. EU AI Act

Correct Answer: D

Explanation: The EU AI Act is the first comprehensive AI-specific law. GDPR is also EU-based but focuses on personal data privacy; the U.S. Blueprint is a set of principles rather than enforceable law; GPAI is an international coalition for best practices.

Q4 [Type: MCQ | Difficulty: Easy | QID : 54959]

When AI is used to cause harm, it is usually because the AI itself has developed bad intentions and gone rogue against its creators.

- A. True
- B. False

Correct Answer: B (False)

Explanation: AI has no intentions - good or bad. It is a tool, like a hammer. Misuse is almost always about humans using AI to harm other humans; the technology amplifies the intent of the person holding it. Safety failures are similarly about unintended consequences of human design choices, not AI "going rogue." This is a movie misconception, not a real risk pattern.

Q5 [Type: MCQ | Difficulty: Medium | QID : 54961]

Even as detection tools improve, deepfake generation tools also improve - so human judgment, skepticism, and verification remain essential to addressing deepfakes, not just technology alone.

- A. True
- B. False

Correct Answer: A (True)

Explanation: This is the cat-and-mouse problem directly: "As detection improves, so does generation." Each generation of detection trains the next generation of fakes. The long-term answer is a combination: better detection tools, source provenance and watermarking, and an informed population that pauses to verify rather than instantly believing what they see. Technology alone will not solve deepfakes.

Q6 [Type: MCQ | Difficulty: Medium | QID : 54965]

Facial recognition data is essentially like a password: if it gets stolen or leaked, you can simply reset it the same way you would reset a compromised account password.

- A. True
- B. False

Correct Answer: B (False)

Explanation: Your facial signature can be stolen - but unlike a password, you cannot reset your face. The same applies to fingerprints, iris scans, and voiceprints - biometric data is permanent. Once compromised, it stays compromised, which is why biometric privacy gets special attention in policy and regulation. A leaked password is annoying; a leaked biometric is forever.

Q7 [Type: MCQ | Difficulty: Easy | QID : 54967]

Which of the following is a form of AI MISUSE (deliberate human use of AI to cause harm)?

- A. A self-driving car fails to detect a pedestrian in unusual lighting conditions on a poorly lit street at night
- B. A chatbot is accidentally trained on flawed help documents that contained errors no one reviewed before deployment
- C. A facial recognition system is designed to work better in good lighting but performs poorly when used at night
- D. Voice cloning used to impersonate a person and trick a victim into transferring money

Correct Answer: D

Explanation: Fraud and impersonation via voice cloning is one of the four named forms of misuse, alongside disinformation at scale, cyber attacks, and harassment / exploitation. The other options are unintended failures or design choices - not deliberate harm.

Q8 [Type: MCQ | Difficulty: Easy | QID : 54970]

Which of the following is a deepfake DETECTION technique?

- A. Voiceprint cloning
- C. Disinformation generation
- D. Predictive maintenance
- D. Blink rate analysis

Correct Answer: D

Explanation: The four detection techniques are blink rate analysis, lighting / shadow inconsistencies, heartbeat detection (subtle skin colour changes from blood flow), and audio-visual mismatch (lip-sync issues). All four exploit small physical signals that current generative models often get wrong.

Q9 [Type: MCQ | Difficulty: Medium | QID : 54973]

A self-driving delivery robot fails to recognise a person in a wheelchair as a pedestrian because the training data contained mostly standing adults. The robot does not stop. Which of the following best describes this incident?

- A. AI Misuse - the company deliberately programmed the robot to ignore people in wheelchairs as part of a targeted exclusion policy
- B. Technical Glitch - the robot's cameras malfunctioned at the moment of the incident
- C. AI Safety Failure - the AI caused unintended harm because the training data lacked wheelchair users
- D. Operator Error - the human supervisor failed to override the robot in time

Correct Answer: C

Explanation: Misuse requires deliberate human intent to cause harm. Here, the harm is real but unintended - caused by training data that did not represent the full range of pedestrians. This is a safety failure: a flaw in design and testing, not a deliberate act.

Q10 [Type: MCQ | Difficulty: Medium | QID : 54976]

Which of the following is a category of personal data at risk from AI systems?

- A. Network packet timing
- B. Behavioural data
- C. Stack trace logs
- D. Image resolution data

Correct Answer: B

Explanation: The five named at-risk categories are voice, facial, behavioural, location, and biometric data. Behavioural data is among the most pervasive because it is collected silently from everyday digital activity - and AI can use it to infer mood, interests, and intent.

Q11 [Type: MCQ | Difficulty: Hard | QID : 54978]

A company's finance team is worried about voice-deepfake fraud after reading about a CEO who was tricked into wiring €220,000 based on a fake phone call. Which safeguard most directly addresses this specific threat?

- A. Multi-factor verification - confirming any large transfer through an independent second channel before money moves
- B. Increasing the volume of marketing emails the company sends to raise general awareness about online fraud risks
- C. Speeding up wire transfers so attackers cannot catch up to the company's payment processing systems
- D. Removing all phones from the office to eliminate any possibility of voice-based communication with finance staff

Correct Answer: A

Explanation: Voice deepfakes work by exploiting trust in a single fast channel (the phone call). The safeguard that breaks the attack is requiring confirmation through a separate channel the attacker has not spoofed - which is the industry-standard recommendation for high-value transfers.

Q12 [Type: MCQ | Difficulty: Medium | QID : 54981]

AI "amplifies" privacy risks in several ways. Which of the following best illustrates one of those amplifying mechanisms?

- A.** AI only stores what users explicitly type into a search box and ignores all other behavioural signals or background activity
- B.** AI deletes all of its data automatically every 24 hours and cannot retain information about users beyond a single day
- C.** AI cannot infer anything beyond what is directly given to it and is unable to combine separate inputs into new conclusions
- D.** AI can combine separate datasets to reveal sensitive details a person never shared - for example, inferring health from shopping and location data

Correct Answer: D

Explanation: "Connecting the Dots" and "Inference Without Consent" are the two clearest amplifiers. A piece of data feels harmless in isolation, but AI can combine it with other data to reach conclusions you never agreed to share - and the opacity of AI systems means you may never know what was inferred about you.

Q13 [Type: MCQ | Difficulty: Medium | QID : 54983]

Why does biometric data - such as facial maps, fingerprints, or voiceprints - pose a unique privacy risk compared to a traditional password?

- A.** Biometric data is only used by government agencies for identification and is never collected by private companies
- B.** Biometric data is much easier for AI systems to guess randomly than complex alphanumeric passwords
- C.** Unlike a password, you cannot reset or change your biological features once they are stolen
- D.** Passwords are more complex for AI systems to analyse than biometric data like fingerprints or voiceprints

Correct Answer: C

Explanation: Biometric data is permanent to your identity. If your facial signature or voiceprint is compromised, you cannot "reset" it the way you would reset a password. This permanence is exactly why privacy regulations treat biometric data as especially sensitive.

Q14 [Type: MCQ | Difficulty: Medium | QID : 55014]

What is the core purpose of the "Human-in-the-Loop" principle in AI governance?

- A.** To reduce the cost of developing AI by using human labour instead of processors
- B.** To ensure humans review and take responsibility for high-stakes AI decisions
- C.** To ensure AI systems eventually learn to operate without any human oversight
- D.** To have humans perform mathematical calculations that the AI finds too difficult

Correct Answer: B

Explanation: AI lacks the ability to understand context or weigh ethical consequences. Human-in-the-Loop ensures that for decisions that affect someone's freedom, health, or livelihood, a human applies judgment and is accountable for the outcome - not the AI.

Q15 [Type: MCQ | Difficulty: Medium | QID : 55019]

Which framework gives individuals rights over their personal data - including the right to know what data is collected, the right to have it deleted, and the right to object to automated decisions?

- A.** The EU AI Act
- B.** The Global Partnership on AI
- C.** The US AI Bill of Rights
- D.** GDPR

Correct Answer: D

Explanation: GDPR (the General Data Protection Regulation) is the specific EU law that grants individuals control over their personal data. The EU AI Act is broader AI legislation; the U.S. Blueprint is a set of principles; GPAI is an international coalition for best practices.

Q16 [Type: MCQ | Difficulty: Medium | QID : 55022]

How do deepfakes create an "Erosion of Trust" - a harm that affects society even when a particular piece of media is actually real?

- A.** By allowing people to dismiss genuine evidence as "just a deepfake" to avoid accountability
- B.** By automatically deleting real videos from the internet so that only manipulated versions remain available
- C.** By making real videos appear in lower resolution and quality compared to AI-generated versions of the same content
- D.** By making it impossible for real cameras to record audio properly during important news events and interviews

Correct Answer: A

Explanation: Once fake media becomes indistinguishable from real, people can use the existence of deepfakes as a shield: claiming a true recording is "just a deepfake" to escape accountability. This is why the harm is bigger than any single fake - it weakens trust in all media, including the genuine kind.

Q17 [Type: MCQ Scenario | Difficulty: Medium | QID : 55025]

Scenario: In 2019, the CEO of a UK-based energy company received a phone call from what sounded exactly like his German boss - same voice, same tone, even the slight German accent. The caller urgently asked him to transfer €220,000 to a Hungarian supplier. The CEO complied. The money was routed through Hungary and Mexico and disappeared. His boss had never made the call.

Question: Which AI technology was used, and what made the case so hard to investigate?

- A.** Computer Vision; the case was hard because the office cameras were turned off and could not capture the suspect's identity
- B.** Robotics; the case was hard because robots are unfamiliar to courts and judges have little experience handling such cases
- C.** AI-powered voice synthesis (an audio deepfake); the criminals were never identified and audio evidence is hard to trace
- D.** OCR; the case was hard because the documents involved were handwritten and could not be reliably scanned by the system

Correct Answer: C

Explanation: The technology was AI voice synthesis trained on the boss's public recordings. The case is hard to solve because no physical evidence ties the attackers to the act, audio alone is hard to attribute, and once money moves through multiple international accounts, recovery becomes nearly impossible.

Q18 [Type: MCQ Scenario | Difficulty: Hard | QID : 55027]

Scenario: A criminal scrapes online videos of a child to capture their voice. The criminal then uses AI to clone that voice and calls the child's parent, mimicking the child and claiming to be in trouble - demanding money urgently.

Question: Which concern (or concerns) does this raise?

- A.** Deepfake concern only - only the cloned audio matters because the videos were already public online, so there is no real privacy violation involved
- B.** Both - the cloned audio is a voice deepfake used for fraud, the child's voice was scraped without consent
- C.** Privacy concern only - only the scraped videos of the child from online sources matter in this case
- D.** Neither - this is just ordinary fraud involving a phone call and a request for money

Correct Answer: B

Explanation: Two harms stack: the deepfake (voice cloning used to deceive and defraud) and the privacy violation (the source recordings were taken without consent). The lesson is that deepfakes and privacy risks often appear together, especially in voice fraud.

Q19 [Type: FRQ - Short Answer | Difficulty: Hard | QID : 55035]

Scenario: A student uses an AI voice-cloning tool to create a fake recording of a classmate saying threatening things about the school, then shares it with the school principal.

Question: Identify whether this is AI Misuse or an AI Safety Failure, justify your classification, AND describe TWO concrete safeguards (technical or policy) that could help prevent similar incidents in the future.

(Respond in 3–4 sentences.)

Model Answer: This is clearly AI Misuse, not a safety failure: the student deliberately used the voice-cloning tool to harass and deceive, which is exactly the kind of intentional harm classified as misuse. The first safeguard is at the platform level - voice-cloning tools should require identity verification and watermark all generated audio so that fabricated recordings can be traced and detected. The second safeguard is at the school level - staff who receive recordings, threats, or accusations should follow a verification protocol (independent confirmation, audio analysis tools, or speaking with the accused student directly) before acting on a single piece of audio evidence. Together these reflect the point that human verification combined with technical safeguards is the best defence against deliberate AI misuse.

Q20 [Type: FRQ - Scenario Analysis | Difficulty: Hard | QID : 55037]

Scenario: A new AI-powered children's toy listens to a child speak, learns their voice, responds in personalised conversation, and uploads conversation snippets to the company's servers for "product improvement." The toy is marketed as fun and educational but has not been independently audited.

Question: Identify at least TWO distinct concerns from this material that this product raises (e.g., privacy, deepfake risk, governance gap), AND propose at least TWO concrete safeguards - referring to ideas such as Human-in-the-Loop, regulation (EU AI Act / GDPR style), and transparency.

(Respond in 3–4 sentences.)

Model Answer: The first concern is privacy: the toy collects voice data - biometric data which is permanent and cannot be "reset" - and uploads it to company servers without clear consent, especially from minors. The second concern is deepfake and fraud risk: stored voice recordings of a child could be misused or leaked to clone the child's voice for impersonation attacks, similar to the AI ransom-call scenario discussed in class. As a first safeguard, the product should comply with privacy regulation (GDPR-style data minimisation, parental consent, and the right to delete recordings) and be transparent about exactly what is collected, stored, and used. As a second safeguard, an independent audit and Human-in-the-Loop oversight should be required before such a product is sold for use with children, possibly under a high-risk classification similar to the EU AI Act, so the level of scrutiny matches the sensitivity of the users.